SERANGOON JUNIOR COLLEGE

General Certificate of Education Advanced Level

Higher 2

CHEMISTRY	**9647/03**
Preliminary Examination	**17 August 2011**
Paper 3 Free Response Questions	**2 hours**

Additional Materials: Data Booklet
Writing Papers

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough work.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in the brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together with the cover page provided.

Answer any **four** questions.

1 Acrylamide is a compound that is found in many deep fried foods such as French fries and potato chips. Lately, it has been added to the list of substances of very high concern in 2010 as it is believed to be carcinogenic.

(a) Acrylamide contains 50.7% by mass of carbon, 7.0% by mass of hydrogen, 19.8% by mass of nitrogen and 22.5% by mass of oxygen. Given that the relative molecular mass of acrylamide is 71.0, determine the molecular formula of acrylamide. Hence, draw a possible structure of acrylamide.

[3]

(b) Apart from deep fried food, acrylamide can also be found in beverages such as prune juice and brewed coffee. Studies have shown that a person can safely ingest up to 0.5 mg / kg body weight in a day without any observed adverse effects to his or her nervous system. Given that 1 kg of brewed coffee contains 1.2×10^{-7} mol of acrylamide, calculate the maximum volume of coffee a 50 kg woman can drink per day. You may assume that coffee has the same density as water.

[3]

(c) State the pH of the solution formed when acrylamide dissolves in water. With reference to the structure of acrylamide, explain your answer clearly.

[3]

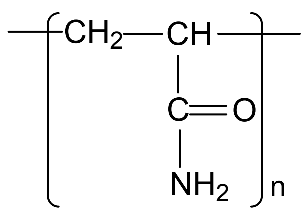
(d) Acrylamide is produced through the hydrolysis of acrylonitrile. This reaction is catalysed by the enzyme nitrile hydratase.

(i) In the process of producing acrylamide, the pH and temperature of the system must be carefully maintained so that the enzyme will not undergo denaturation. Explain what is meant by the term *denaturation*.

(ii) Explain in terms of structure and bonding how extreme pH and temperature can cause denaturation of an enzyme.

[5]

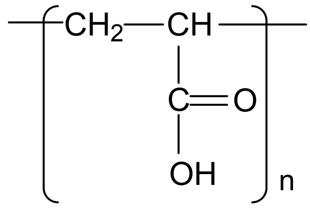
(e) When molecules of acrylamide are heated, they undergo polymerisation to form the polymer, polyacrylamide. The structure of polyacrylamide is shown below:



(i) Polyacrylamide is sometimes used in disposable diapers as it is able to absorb as many times its mass, of water. However, when sodium chloride is added to the polymer, the absorbed water is immediately released.
Explain.

(ii) Aqueous solutions of polyacrylamide are found to undergo chemical degradation readily when pH and temperature are high. Identify the type of reaction the polymer has undergone and describe the products formed.

(iii) Polyacrylic acid is a polymer that is quite similar to polyacrylamide. The structure of polyacrylic acid is shown below:



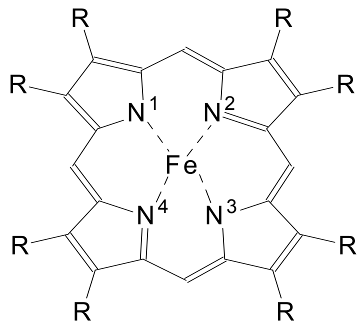
Suggest how polyacrylamide can be distinguished from polyacrylic acid, without the use of any strong mineral acids or bases.

[6]

[Total: 20]

2 Metalloproteins is a generic term for a protein that contains a metal ion which is required to assist in the protein's biological activity. Three metalloproteins that can be found in ocean organisms are the iron-based haemoglobin, copper-based haemocyanin and the vanadium-based vanabin.

(a) Haemoglobin has a quaternary structure that contains globular proteins and embedded haem groups. The diagram shows a simplified version of the haem group. The various different side-groups are all shown as R.



Note that the four N atoms and the Fe ion are planar.

(i) Suggest what is meant by the term *ligand*.

(ii) State the difference between the type of bonds that occur between Fe ion and the numbered N^1 and N^2 atoms.

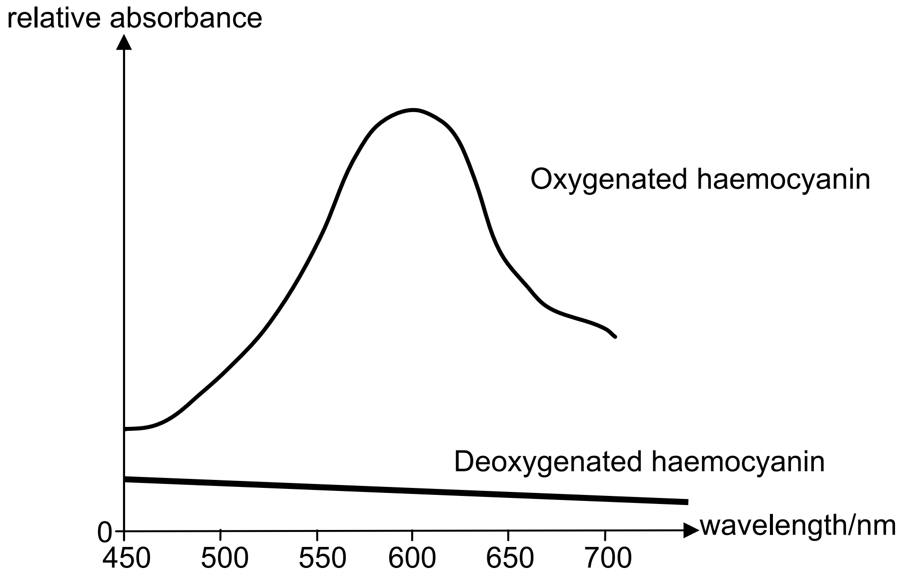
(iii) One molecule of haemoglobin (Hb) can bind up to four molecules of oxygen, according to the following equation.

$$Hb(aq) + 4O_2(aq) \rightleftharpoons Hb(O_2)_4(aq)$$

By considering the role of oxygen in this equation, comment on how the body cells receive oxygen from the air in the lungs.

[5]

(b) The visible absorption spectra of the oxygenated and deoxygenated forms of haemocyanin are shown below:



(i) What are the colours of the two forms of haemocyanin?

(ii) Explain why the oxygenated haemocyanin is coloured.

(iii) State the full electronic configuration of the copper ion in the deoxygenated haemocyanin.

(iv) Suggest the wavelength of the absorption peak for a sample of red-pigmented haemoglobin.

[6]

(c) Vanabin in sea squirts accumulates vadadium in the blood cell by using nicotinamide adenine dinucleotide phosphate (or NADPH) as a reducing agent to give the sea squirts their distinct green colouration.

$NADP^+ (aq) + H^+ (aq) + 2e^- \rightleftharpoons NADPH (aq)$ $E^\ominus$ –0.32V

Use the standard redox potential data in *Data Booklet* and writing equations wherever possible to predict the outcome of the reactions of vanadate (V) ions with excess NADPH. Hence, determine the ion that is responsible for the green colouration in sea squirts.

[4]

(d) Group II elements are also being involved in metalloproteins. An example is chlorophyll. Chlorophyll is vital for the process of photosynthesis, and it contains a magnesium ion. Unlike haemoglobin, which utilises the variable oxidation state of iron ion to transport oxygen, chlorophyll makes use of another mechanism to fulfil its function.

(i) Account for the tendency for transition metals ions to have variable oxidation states.

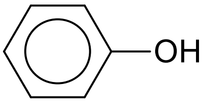
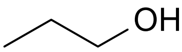
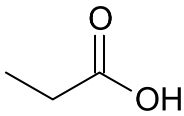
(ii) Explain why transition metals carbonates such as copper (II) carbonate often decompose at a lower temperature than Group II carbonates.

(iii) Discuss why the densities of the first transition element series increase along the sequence as far as copper.

[5]

[Total: 20]

3 The labels to three bottles of organic compounds each containing either

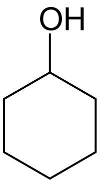
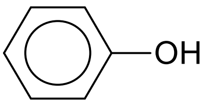
 ,  or  had been stained such that only their pK_a values are visible.

The pK_a values on these 3 bottles read 4.87, 9.95 and 16.0.

(a) **(i)** Explain what is meant by *acid dissociation constant*.

(ii) Assign the correct pK_a value to the 3 compounds and explain your answer.

(iii) Describe how you can carry out a simple chemical test to distinguish between

 and 

[8]

(b) Buffered propanoic acid mixes have always been used as preservatives in the process of growing *alfalfa-grass*. It is by far, the most effective in inhibiting growth of moulds. To ensure its effectiveness, the propanoic acid/sodium propanoate buffer system is maintained at pH 6.

(i) With the aid of equation(s) or otherwise, explain how the buffer system helps to maintain pH.

(ii) Using the pKa value of propanoic acid that you have identified in **(a)(ii)**, calculate the ratio of propanoate and propanoic acid in the buffer system.
Hence, determine the new pH of a 2 dm^3 buffer solution containing 0.05 mol dm^{-3} of propanoic acid when 0.03 mol of H^+ is added.

[6]

(c) An organic compound **A**, $C_5H_8O_2$, gives a weakly acidic solution in water. When hydrogen bromide gas was reacted with **A**, a chiral compound **B**, $C_5H_9O_2Br$ was produced. On heating **B** with aqueous NaOH followed by concentrated H_2SO_4, a sweet smelling neutral liquid **C** yielded. Compound **C** has the same molecular formula as **A**. When 1 mol of **A** was heated with $KMnO_4$ in dilute H_2SO_4, 1 mol of an acidic gas was produced.

Deduce the structural formula of **A**, **B**, and **C.**

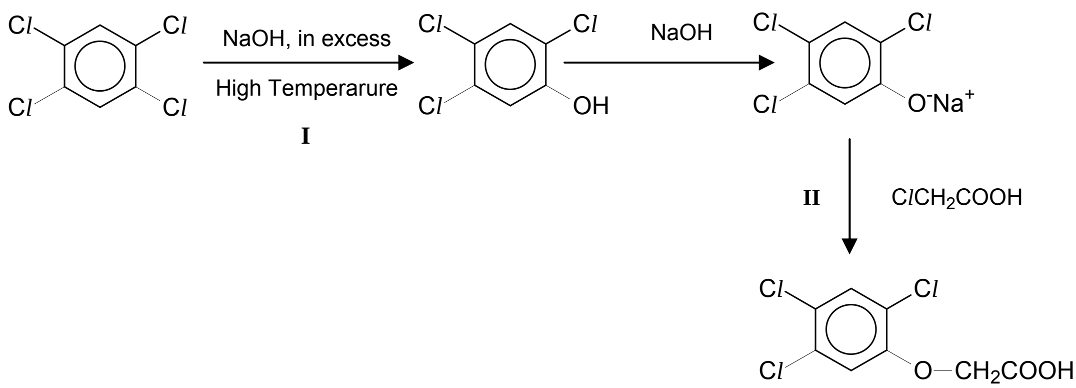
[6]

[Total: 20]

4 Herbicides, also known as weedkillers are widely used in agriculture.

2,4,5-Trichlorophenoxyacetic acid (2,4,5-T) was a widely used herbicide until being phased out starting in the late 1970s due to health reasons.

(a) 2,4,5-T can be synthesised by the following process:



(i) What is unusual about reaction **I**? Explain your answer.

(ii) What type of reaction is reaction **II**? Describe the mechanism involved.

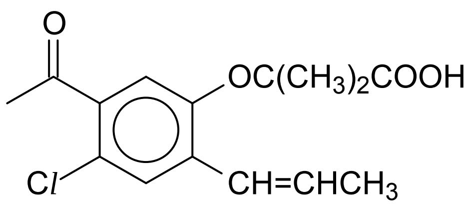
[5]

(b) To test for presence of halogenoalkanes, silver nitrate is often used to precipitate out any halide ions present after reactions.
An example of such a precipitate is silver chloride. The solubility product of silver chloride at a given temperature is 1.6×10^{-5} mol^2 dm^{-6}.

(i) Calculate the solubility of silver chloride in water.
(ii) Calculate the minimum concentration of dilute hydrochloric acid needed to precipitate out silver chloride if 2 cm^3 of the acid is added to 10 cm^3 of 0.1 mol dm^{-3} of silver nitrate.
(iii) Explain what happens when aqueous hydrochloric acid is added to a solution of silver nitrate followed by aqueous ammonia in excess.

[6]

(c) Another common household weed killer, *mecoprop* has the following structure:



(i) What type of isomerism is displayed by *mecoprop*?

(ii) Draw the structure of the product formed when *mecoprop* reacts with

1 PCl_5

2 2,4- DNPH [3]

(d) Compound **D**, C_9H_8O, insoluble in water and aqueous sodium hydroxide, reacts with chlorine gas in the dark to form **E**, $C_9H_8OCl_2$. **D** does not react with Fehling's solution but reacts with Tollens' reagent to give a silver mirror. On heating with acidified potassium manganate(VII), **D** produces **F**, $C_8H_6O_4$, which is soluble in sodium hydroxide.
Deduce the structures of compounds **D**, **E**, and **F**.

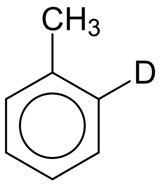
[6]

[Total: 20]

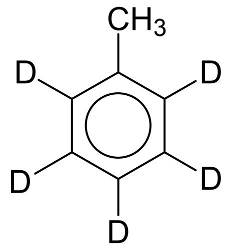
5 Deuterium (D or ^2H) is a heavy isotope of hydrogen.

Chlorine can react with deuterium to form deuterium chloride, DC*l*.

(a) When DC*l* is dissolved in methylbenzene, no reaction occurs. However, if an anhydrous catalyst **G** is added to the solution, a reaction takes place, and a compound **H** is one of the products. If more DC*l* is present, a compound **I** is formed.



H

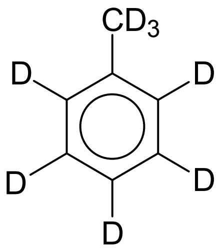


I

(i) Explain why **G** can act as the catalyst in the reaction.

(ii) Suggest a suitable identity for the catalyst **G**.

(iii) In the presence of *excess* DC*l*, would compound **J** be formed? Explain. [4]



J

(b) Propanone, known industrially as acetone, is an important industrial solvent. It is an active ingredient in nail polish remover.

The bromination of propanone is acid-catalysed:

$$CH_3COCH_3 + Br_2 \xrightarrow{H^+} CH_3COCH_2Br + H^+ + Br^-$$

The rate of disappearance of bromine was measured for several different concentrations of propanone, bromine and hydrogen ions at a certain temperature.

Expt	$[CH_3COCH_3]$ /mol dm^{-3}	$[Br_2]$ /mol dm^{-3}	$[H^+]$ /mol dm^{-3}	Rate / mol dm^{-3} s^{-1}
1	0.30	0.05	0.05	5.7×10^{-5}
2	0.30	0.10	0.05	5.7×10^{-5}
3	0.30	0.05	0.10	1.2×10^{-4}
4	0.40	0.05	0.20	3.1×10^{-4}
5	0.40	0.05	0.05	7.6×10^{-5}

(i) Determine the order of reaction with respect to each reactant and hence the rate equation.

(ii) In another experiment, the concentration of CH_3COCH_3 used was in excess. As a result, a new rate law was established as follows:

Rate = k' [H^+]

If the initial concentration of H^+ ions is 1.6 mol dm^{-3} and the half-life for the reaction is 5 minutes, sketch a graph of concentration of H^+ against time for the first 20 minutes of the reaction.

(iii) The following steps in a mechanism are as shown:

K + **L** $\longrightarrow$ **KL** (slow)

KL + **M** $\longrightarrow$ **N** (fast)

Given that the activation energy of the reaction is +35 kJ mol^{-1} and ΔH is -90 kJ mol^{-1}, sketch the energy profile diagram for the above reaction.

(iv) Using an appropriate diagram, explain how higher temperatures affect the rate of reaction.

(v) A quarter of the world's propanone is used to make methyl methacrylate, $CH_2=C(CH_3)CO_2CH_3$ which is used on a large scale to produce plastics. The first step is to convert propanone to its corresponding cyanohydrin. The cyanohydrin is then further reacted to form methyl methacrylate.
State the reagents and conditions required to convert propanone to its corresponding cyanohydrin. Draw the structure of the cyanohydrin.

[13]

(c) Organometallic compounds, usually a metal attached to an R group, can be used to convert carbonyl compounds to alcohols. An example is shown below:

$$R{-}CH{=}O \ + \ R'{-}Li \xrightarrow{H_2O} R{-}CH(OH){-}R' \ + LiOH$$

2-bromobutane can be converted to 2-methylbutan-2-ol in **3 steps**. Write the reagents and conditions required for the conversion, including intermediate compounds.

[3]

[Total: 20]

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